CSA06507-Generative AI

Lab Experiment's

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Unit 3: Lab Programs for Prompting

Experiment 2: Develop a Semantic Search System Using Cosine Similarity Between Document and Query Embedding

Aim:

To develop a semantic search system that converts documents and user queries into numerical embeddings and calculates cosine similarity to retrieve the most relevant documents based on their meaning.

Algorithm

1. Start the program.
2. Import the required Python libraries.
3. Create a collection of sample documents.
4. Load the Sentence Transformer model.
5. Convert all documents into numerical embeddings.
6. Accept a search query from the user.
7. Convert the query into an embedding.
8. Calculate cosine similarity between the query and document embeddings.
9. Sort the documents based on similarity scores.
10. Display the most relevant documents.
11. Stop the program.

Python Program

Google Colab – Step 1: Install required libraries

Python

Run

```
!pip install sentence-transformers scikit-learn
```

Step 2: Python Code

Copy and run this code in another Colab cell.

Python

Run

```
# Semantic Search Using Cosine Similarity
```

```
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity

# Step 1: Load the model
model = SentenceTransformer('all-MiniLM-L6-v2')

# Step 2: Create sample documents
documents = [
    "Artificial Intelligence helps machines learn and solve problems.",
    "Python is a popular programming language.",
    "Machine Learning is a branch of Artificial Intelligence.",
    "Computer networks connect multiple computers.",
    "Deep Learning uses neural networks to process data."
]

# Step 3: Generate document embeddings
doc_embeddings = model.encode(documents)

# Step 4: Get user query
query = input("Enter your search query: ")

# Step 5: Generate query embedding
query_embedding = model.encode([query])

# Step 6: Calculate cosine similarity
similarity_scores = cosine_similarity(
    query_embedding,
    doc_embeddings
)[0]

# Step 7: Rank documents
results = sorted(
```

zip(documents, similarity_scores),

key=lambda x: x[1],

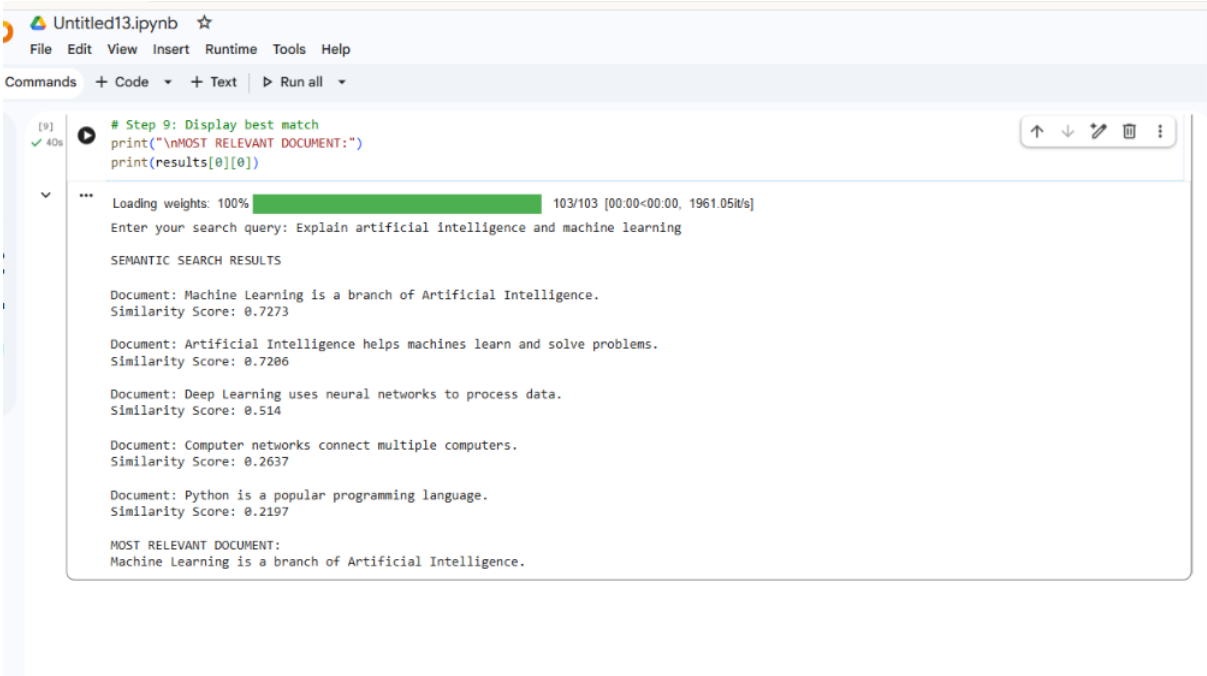
Input

Example input:

Plain text

Enter your search query: Explain artificial intelligence and machine learning

Output



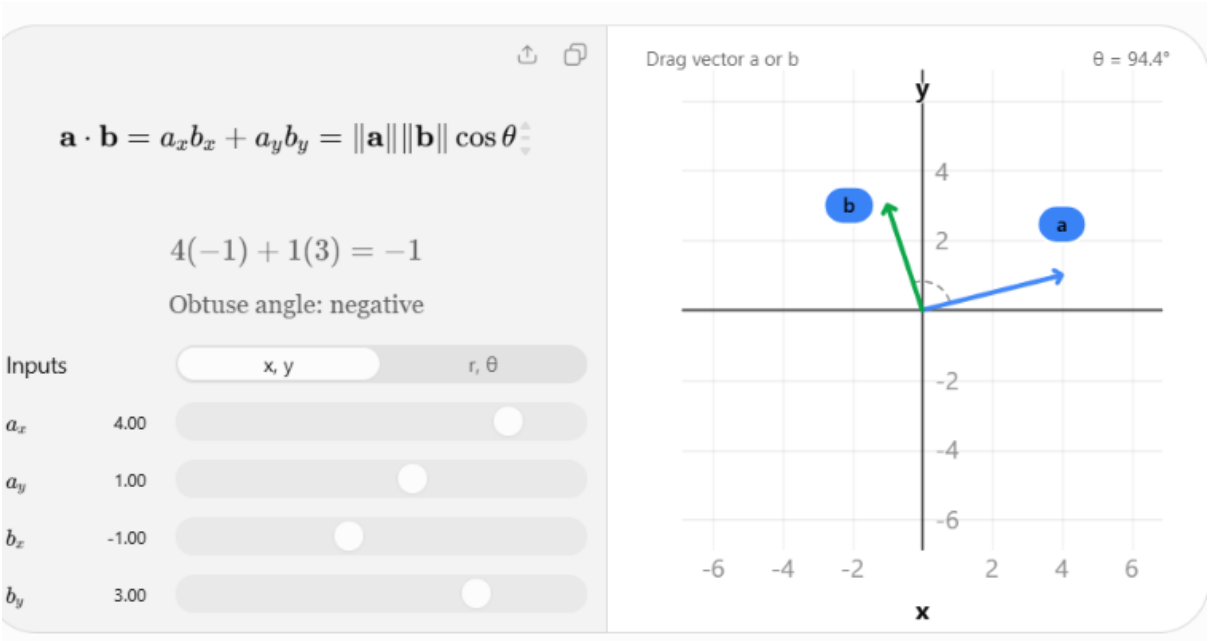
Explanation

1. Document Embedding: Converts the stored documents into numerical vectors.

2. Query Embedding: Converts the user's search query into a numerical vector.

3. Cosine Similarity: Measures the similarity between the query vector and document vectors.

4. Ranking: Sorts the documents according to their similarity scores. Higher scores indicate greater similarity.



Result

Thus, a semantic search system was successfully developed using Python and Google Colab. Document and query embeddings were generated using the Sentence Transformer model. Cosine similarity was calculated to identify and rank the most relevant documents based on the meaning of the user's query.